

Motov - MotorBike rental management system

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Abstract. The rapid growth of digital platforms has transformed the transportation and rental industry. However, many motorbike rental businesses still rely on manual processes, resulting in inefficient booking management, vehicle tracking, identity verification, and payment handling. This paper proposes Motov, a hybrid motorbike rental management platform that integrates company-owned and owner-owned vehicles into a unified ecosystem. The system provides web and mobile applications supporting customers, staff, vehicle owners, and administrators. Core functionalities include vehicle catalog management, online booking, rental conflict validation, identity verification, payment processing, notification services, and administrative analytics. The proposed solution aims to improve operational efficiency, increase vehicle utilization, enhance customer experience, and support secure rental transactions. The platform is implemented using ReactJS, React Native, ASP.NET Core Web API, and MongoDB.

Keywords: Motorbike Rental · Vehicle Management · Online Booking · Identity Verification · Mobile Application

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1 Introduction

The demand for motorbike rental services has increased significantly due to tourism growth, urban mobility needs, and the popularity of short-term transportation solutions. In Vietnam, many rental businesses still operate through manual processes such as phone calls, social media messaging, and paper-based agreements.

Traditional rental approaches often face several challenges, including inefficient vehicle management, limited booking transparency, and difficulties in tracking rental transactions. Furthermore, customers may experience inconvenience when searching for available vehicles, while rental businesses struggle to maximize vehicle utilization.

Recent studies have investigated vehicle rental systems, fleet management, customer satisfaction, and peer-to-peer vehicle sharing models [3, 2, 5, 4, 6, 7].

This project proposes Motov, a hybrid motorbike rental platform that supports both company-owned and partner-owned vehicles. The system allows customers to search and book vehicles online, partners to register and manage their vehicles, and administrators to oversee platform operations.

The main contributions of this project are:

- Development of a hybrid rental model integrating company-owned and owner-owned motorbikes.
- Design of a centralized rental management platform supporting customers, staff, owners, and administrators.
- Integration of identity verification, booking conflict validation, and automated rental price calculation mechanisms.
- Implementation of web and mobile applications using modern full-stack technologies.
- Improvement of booking transparency, operational efficiency, and vehicle utilization.

2 Related Work

Several studies have investigated vehicle rental, fleet management, customer satisfaction, and car-sharing systems.

Khawaja and Yang [3] examined the relationship between employee engagement, customer satisfaction, and customer retention in the car rental industry. Their findings indicated that service quality significantly influences customer loyalty and business performance. However, the study focused on customer behavior rather than software platform development.

Gonzalez et al. [2] proposed a fleet utilization metric for shared mobility systems. Although the study improved operational evaluation, it did not address booking management or payment integration.

Munzel et al. [5] analyzed different car-sharing business models and demonstrated the viability of peer-to-peer vehicle sharing. However, the research did not provide a practical software implementation.

Liu et al. [4] investigated booking optimization algorithms for online car-sharing systems. Their work improved vehicle allocation efficiency but lacked user management and payment functionalities.

Neifer et al. [6] explored trust-building mechanisms in peer-to-peer car-sharing platforms. The study highlighted the importance of reputation systems but did not provide a complete rental management solution.

Andersen [1] proposed a queueing model for rental scheduling and inventory optimization. However, the study focused on mathematical optimization rather than platform implementation.

Soppert et al. [7] studied the integration of traditional rental and car-sharing services. Their findings showed improved utilization and profitability, but no web or mobile implementation was proposed.

Although previous studies have focused on fleet utilization, scheduling optimization, customer satisfaction, and peer-to-peer sharing, few studies provide

a comprehensive platform that integrates booking, payment processing, vehicle management, partner vehicle registration, and administrative monitoring.

Therefore, this project proposes a hybrid motorbike rental platform that combines company-owned and partner-owned vehicles through a centralized web and mobile ecosystem.

3 Method

3.1 Development Process

The Agile Scrum methodology was adopted to support iterative development and continuous feedback throughout the project lifecycle. The development process consisted of requirement analysis, sprint planning, implementation, testing, and review activities. This approach allowed the team to incrementally deliver system functionalities while adapting to evolving project requirements.

3.2 System Architecture

The proposed system adopts a three-tier architecture:

- Frontend: ReactJS
- Mobile Application: React Native
- Backend: ASP.NET Core Web API
- Database: MongoDB

3.3 Actors

The proposed system involves five primary actors:

- Guest
- Customer
- Staff
- Owner
- Administrator

3.4 Main Functionalities

The system provides the following functionalities:

- User registration and authentication
- Google Single Sign-On (SSO)
- Profile management and password recovery
- Motorbike browsing, searching, filtering, and sorting
- Vehicle detail and recommendation display
- Favorite vehicle management
- Online booking and rental scheduling

- Vehicle availability checking
- Rental conflict validation
- Rental price calculation
- Online payment and cash payment support
- Citizen ID verification
- Booking tracking and rental history
- Review and rating management
- Notification and reminder services
- Vehicle inventory management
- Promotion management
- Analytics and reporting dashboard

4 System Analysis

4.1 Problem Analysis

Many motorbike rental businesses still depend on manual workflows, including phone-based reservations, social media communication, and paper-based contracts. These approaches often lead to inefficient booking management, poor vehicle utilization, booking schedule conflicts, lack of customer identity verification, and limited operational visibility. In addition, rental businesses face difficulties in monitoring fleet availability, tracking rental transactions, and generating business reports.

4.2 Proposed Solution

The proposed solution is a centralized hybrid motorbike rental platform that supports both company-owned and owner-owned vehicles. The system provides web and mobile applications for customers, owners, staff, and administrators. It integrates online booking, payment processing, citizen identity verification, notification services, rental conflict validation, inventory management, and business analytics into a single ecosystem.

5 System Design

5.1 Use Case Design

Main actors:

- Guest
- Customer
- Owner
- Staff
- Administrator

The use case diagrams for each actor are presented in Figures 1–6.

Figure 7 presents the Entity Relationship Diagram (ERD) of the proposed system.

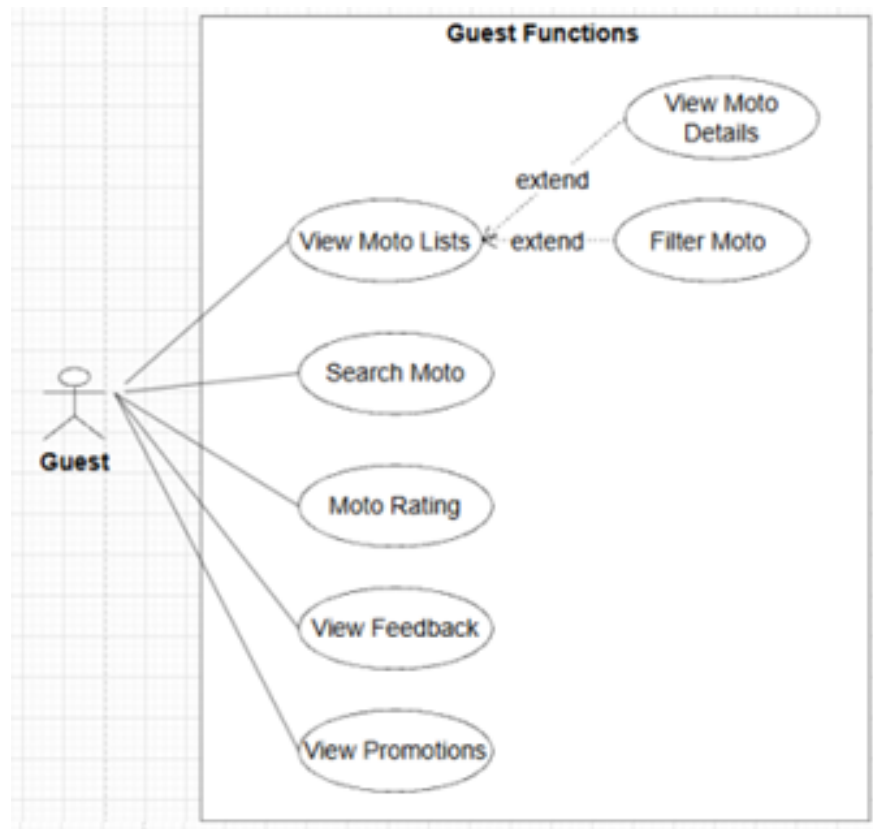


Fig. 1. Guest Use Case Diagram

6 Implementation

6.1 Technologies

- ReactJS
- React Native
- ASP.NET Core
- MongoDB
- GitHub
- Postman

6.2 Current Progress

- Requirements Analysis
- Literature Review
- Use Case Design

- Database Design (ERD)
- User Interface Design
- Authentication Module
- Google Authentication Integration
- Citizen ID Verification Design
- Booking Management Module
- Rental Conflict Validation Logic
- Rental Price Calculation Module
- Payment Integration Design
- Notification Service Design
- CRUD Functions for Users, Vehicles, and Bookings

7 Conclusion

This paper presented Motov, a hybrid motorbike rental platform integrating company-owned and partner-owned vehicles into a unified rental ecosystem. The proposed system supports customers, partners, and administrators through web and mobile applications. Future work includes recommendation systems, real-time vehicle tracking, and advanced analytics.

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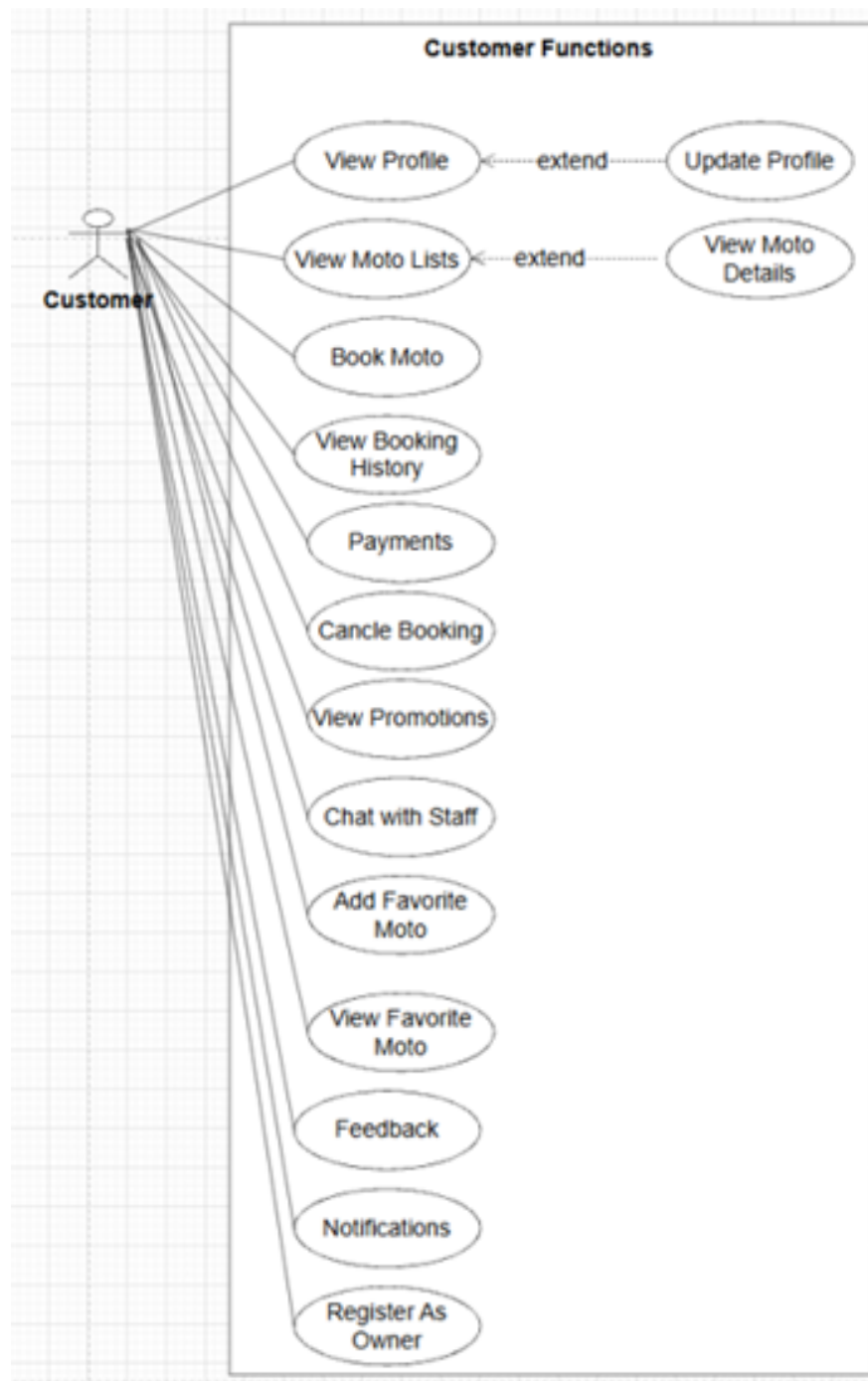
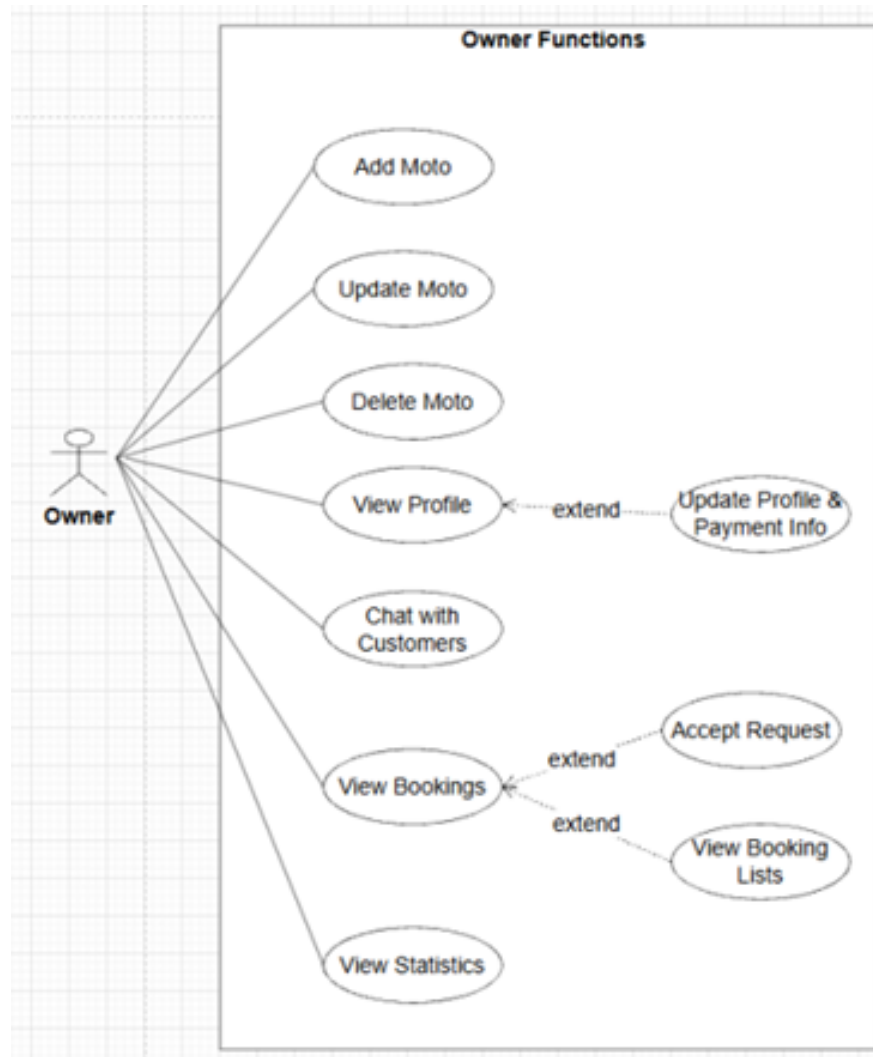


Fig. 2. Customer Use Case Diagram

**Fig. 3.** Owner Use Case Diagram

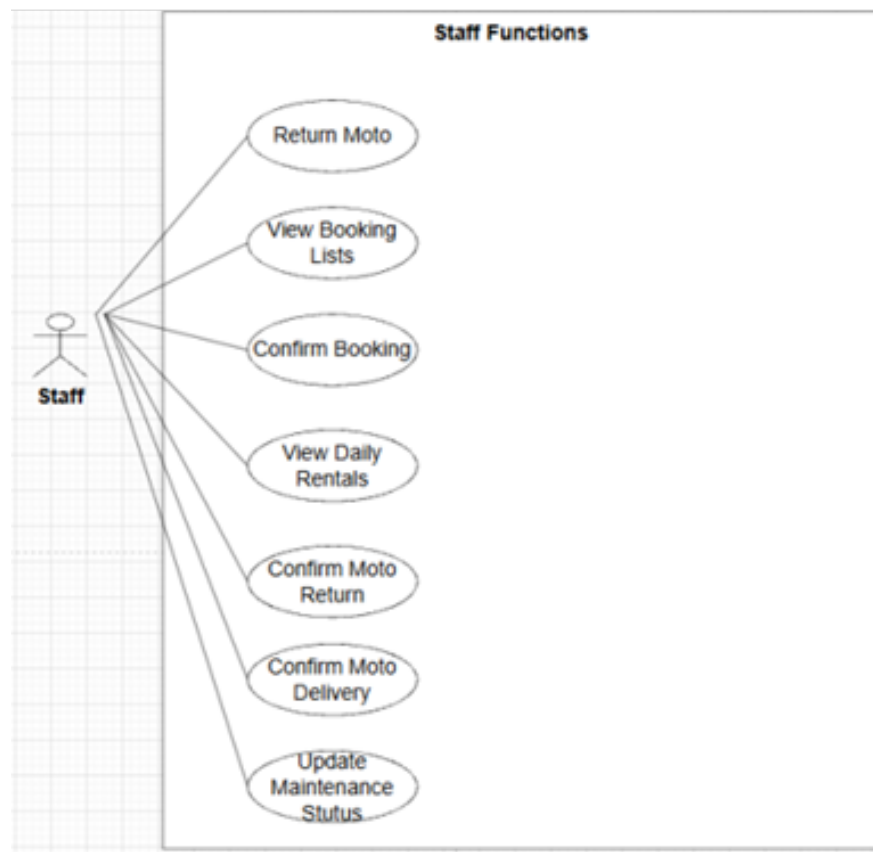


Fig. 4. Staff Use Case Diagram

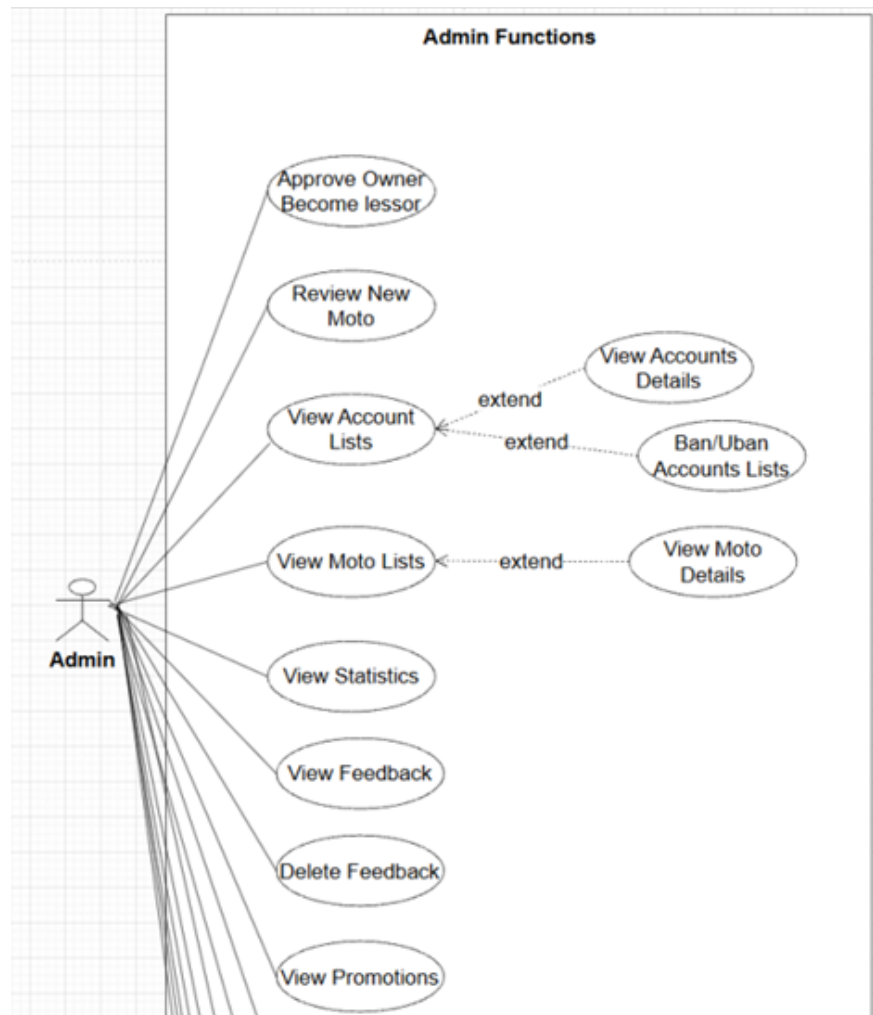


Fig. 5. Administrator Use Case Diagram (Part 1)

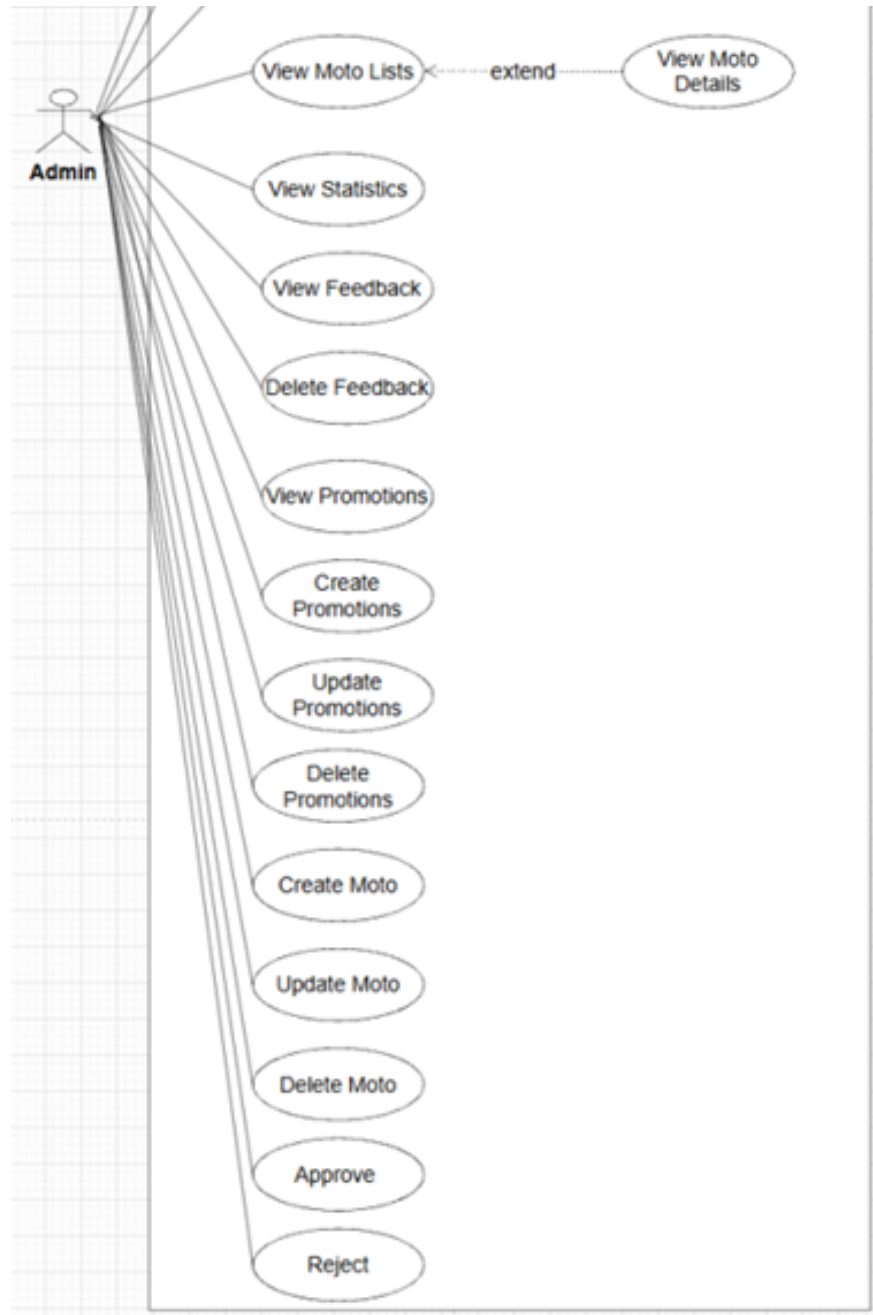


Fig. 6. Administrator Use Case Diagram (Part 2)

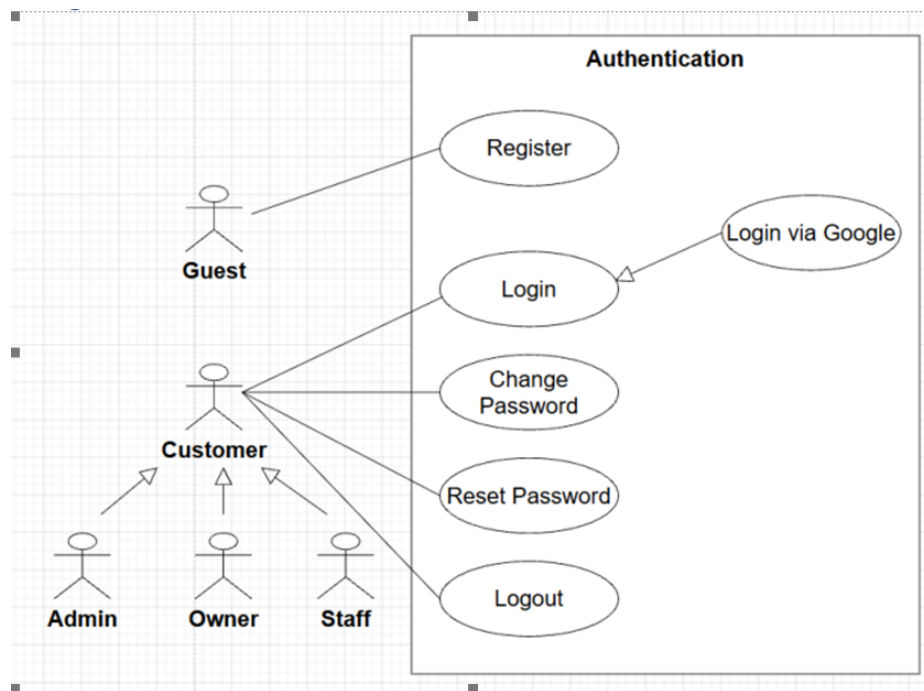


Fig. 7. Entity Relationship Diagram of the Motov system